Computer Science Practical List 5

1. Write a Python code with the following functions performing mentioned operations on a csv file "CANDIDATE.CSV" containing lists with [<Candidate No of type int>,<Cname of type string>,<Score of type float>]
   a. Enrol(), to add details of new CANDIDATEs from the user and save in "CANDIDATE.CSV".
   b. GetPass(), to read content from "CANDIDATE.CSV" and display details of those candidates whose Score is>=50
   c. ShowAll(), to display the entire content of "CANDIDATE.CSV"
   d. AverageScore(), to find and return the average of Score from "CANDIDATE.CSV"

   Also, write menu driven options to call the above functions using a loop.

2. Write a Python code with the following functions performing mentioned operations on a CSV file "ACCOUNTS.CSV" containing lists with [<Ano of type int>,<Name of type string>,<Balance of type float>]
   a. Register(), to add details of new account holders as entered by the user in the CSV file named ACCOUNTS.CSV.
   b. Transact( ), to perform a transaction i.e. Deposit or Withdraw an Amount in an Account Holders Balance whose Acno is entered by the user from ACCOUNTS.CSV.
      Note: an Account Holder must always have a minimum Balance of Rs 500 in his/her Account.
   c. DisplayAll( ) to display the details of all Account Holders from ACCOUNTS.CSV
   d. BankBalance() to find and return the sum of Balance of all the account holders from ACCOUNTS.CSV

   Also, write a menu driven code to call the above functions using a loop.

3. Write a Python code with the following functions performing mentioned operations on a CSV file "EMPLOYEE.CSV" containing lists with [<Eno of type int>,<EName of type string>,<Salary of type float>,<Age of type int>]
   a. Add(), to add details of an employee as entered by the user in the CSV file named EMPLOYEE.CSV.
   b. Delete( ), to delete all those records of employees where the age is above 60 from the file.
   c. Output( ) to display the details of all employees who belong to the "HR" department and whose age is between 40 and 45 (inclusive of both) from the file.
   d. DisplayAll() to display details of all the employees.

   Also, write a menu driven code to call the above functions using a loop.

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN ASSIGNMENT BOOKLET

General Instructions:

i. Type the solution of all the problems using **Python** Language on **Python IDLE** or **Colab**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No      : L5-P1
# Developed By         : Reema Jones
# Date                 : 1-September-2024
# ------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. **"Turn in"** the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy of **each program on separate pages** from the printer and get them signed by the respective computer teacher. Add all the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**10**" for all programs with single line spacing. Default indentation **2 Character** only.

```
# -----------------------------------------------------------------#
# List-Program No        : L5-P1
# Developed By           : Puja Dey
# Date                   : 31-September-2025
# -----------------------------------------------------------------#

import csv
def Enrol():
  with open("CANDIDATE.CSV", "w") as f:
    while True:
            num = int(input("Enter Candidate No: "))
            cname = input("Enter Candidate Name: ")
            score = float(input("Enter Score: "))
            Pen= csv.writer(f)
            Pen.writerow([num, cname, score])
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def ShowAll():
  try:
    with open("CANDIDATE.CSV", "r") as f:
      Records=csv.reader(f)
      for r in Records:
        print(r[0], r[1], r[2])
  except FileNotFoundError:
    print("Candidate.CSV file not Found")

def GetPass():
  try:
    with open("CANDIDATE.CSV", "r") as f:
      Records=csv.reader(f)
      for r in Records:
        if float(r[2]) >=50:
          print(r[0], r[1], r[2])
  except FileNotFoundError:
    print("Candidate.CSV File Not Found")


def AverageScore():
  S=0
  try:
    with open("CANDIDATE.CSV", "r") as f:
      Records=csv.reader(f)
```

```
      n=0
      for r in Records:
        S=S+float(r[2])
        n+=1
      if n > 0:
        print(S/n)
      else:
        print("No records found to calculate average.")
  except FileNotFoundError:
    print("Candidate.CSV File Not Found")

while True:
    print("\n1. Enrol\n2. ShowAll\n3. GetPass\n4. AverageScore\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Enrol()
    elif ch == 2:
        ShowAll()
    elif ch == 3:
        GetPass()
    elif ch == 4:
        AverageScore()
    elif ch == 5:
        break
```

```
1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 1
Enter Candidate No: 101
Enter Candidate Name: Alpha
Enter Score: 30
More records (Y/N)?: Y
Enter Candidate No: 102
Enter Candidate Name: Beta
Enter Score: 50
More records (Y/N)?: Y
Enter Candidate No: 103
Enter Candidate Name: Gamma
Enter Score: 70
More records (Y/N)?: N

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 2
101 Alpha 30.0
```

```
102 Beta 50.0
103 Gamma 70.0

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 3
102 Beta 50.0
103 Gamma 70.0

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 4
50.0

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L5-P2
# Developed By    : Puja Dey
# Date            : 31-September-2025
# ------------------------------------------------------------#
import csv

def Register():
    with open("ACCOUNTS.CSV", "a", newline="") as f:
        w = csv.writer(f)
        while True:
            ano = int(input("Enter Account No: "))
            name = input("Enter Name: ")
            balance = float(input("Enter Balance: "))
            w.writerow([ano, name, balance])
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def DisplayAll():
    try:
        with open("ACCOUNTS.CSV", "r", newline="") as f:
            r = csv.reader(f)
            for row in r:
                if row:
                    print(row[0], row[1], row[2])
    except FileNotFoundError:
        print("No File!")

def Transact():
    try:
        records = []
        acno = int(input("Enter Account No for transaction: "))
        with open("ACCOUNTS.CSV", "r", newline="") as f:
            r = csv.reader(f)
            for row in r:
                if row:
                    ano, name, bal = int(row[0]), row[1], float(row[2])
                    if ano == acno:
                        print("Current Balance:", bal)
                        t = input("Deposit or Withdraw (D/W)?: ").upper()
                        amt = float(input("Enter Amount: "))
                        if t == "D":
                            bal += amt
```

```
                        elif t == "W":
                            if bal - amt >= 500:
                                bal -= amt
                            else:
                                        print("Insufficient balance! Minimum 500 must
    remain.")
                        print("Updated Balance:", bal)
                    records.append([ano, name, bal])
            with open("ACCOUNTS.CSV", "w", newline="") as f:
                w = csv.writer(f)
                w.writerows(records)
        except FileNotFoundError:
            print("No File!")

def BankBalance():
    try:
        total = 0
        with open("ACCOUNTS.CSV", "r", newline="") as f:
            r = csv.reader(f)
            for row in r:
                if row:
                    total += float(row[2])
        print("Total Bank Balance:", total)
    except FileNotFoundError:
        print("No File!")

while True:
    print("\n1. Register")
    print("2. DisplayAll")
    print("3. Transact")
    print("4. BankBalance")
    print("5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Register()
    elif ch == 2:
        DisplayAll()
    elif ch == 3:
        Transact()
    elif ch == 4:
        BankBalance()
    elif ch == 5:
        break
Enter Account No: 102
```

```
Enter Name: Beta
Enter Balance: 200
More records (Y/N)?: Y
Enter Account No: 103
Enter Name: Gamma
Enter Balance: 1000
More records (Y/N)?: N

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 2
101 Alpha 5000.0
102 Beta 200.0
103 Gamma 1000.0

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 3
Enter Account No for transaction: 102
Current Balance: 200.0
Deposit or Withdraw (D/W)?: W
Enter Amount: 100
Insufficient balance! Minimum 500 must remain.
Updated Balance: 200.0

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 3
Enter Account No for transaction: 101
Current Balance: 5000.0
Deposit or Withdraw (D/W)?: D
Enter Amount: 0
Updated Balance: 5000.0

1. Register
```

```
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 4
Total Bank Balance: 6200.0

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L5-P3
# Developed By    : Puja Dey
# Date            : 31-September-2025
# ------------------------------------------------------------#

import csv

def Add():
    with open("EMPLOYEE.CSV","a",newline="") as cF:
        Pen = csv.writer(cF)
        Eno = input("Eno:")
        EName = input("EName:")
        Salary = input("Salary:")
        Age = input("Age:")
        Dept = input("Dept:")
        Pen.writerow([Eno, EName, Salary, Age, Dept])


def Delete():
    try:
        Rows = []
        with open("EMPLOYEE.CSV","r") as cF:
            Rec = csv.reader(cF)
            for r in Rec:
                try:
                    if int(r[3]) <= 60:
                        Rows.append(r)
                except:
                    continue
        with open("EMPLOYEE.CSV","w",newline="") as cF:
            Pen = csv.writer(cF)
            Pen.writerows(Rows)
    except FileNotFoundError:
        print("EMPLOYEE.CSV File Not Found")


def Output():
    try:
        with open("EMPLOYEE.CSV","r") as cF:
            Rec = csv.reader(cF)
            print("HR Department Employees Age 40-45")
            for r in Rec:
                try:
```

```
                    if r[4].upper() == "HR" and 40 <= int(r[3]) <= 45:
                        print(r[0], r[1], r[2], r[3], r[4])
                except:
                    continue
    except FileNotFoundError:
        print("EMPLOYEE.CSV File Not Found")


def DisplayAll():
    try:
        with open("EMPLOYEE.CSV","r") as cF:
            Rec = csv.reader(cF)
            for r in Rec:
                print(r)
    except FileNotFoundError:
        print("EMPLOYEE.CSV File Not Found")


while True:
    OPT = input("A:Add D:Delete O:Output V:ViewAll Q:Quit : ")
    if OPT in ['a','A']:
        Add()
    elif OPT in ['d','D']:
        Delete()
    elif OPT in ['o','O']:
        Output()
    elif OPT in ['v','V']:
        DisplayAll()
    elif OPT in ['q','Q']:
        break
    else:
        print("Invalid Option")
```

```
A:Add D:Delete O:Output V:ViewAll Q:Quit : V
['1', 'Apple', '5000', '32', 'Fruits']
A:Add D:Delete O:Output V:ViewAll Q:Quit : A
Eno:2
EName:Banana
Salary:1000
Age:88
Dept:Yellow
A:Add D:Delete O:Output V:ViewAll Q:Quit : A
Eno:3
EName:Pineapple
```

```
Salary:40000
Age:44
Dept:HR
A:Add D:Delete O:Output V:ViewAll Q:Quit : V
['1', 'Apple', '5000', '32', 'Fruits']
['2', 'Banana', '1000', '88', 'Yellow']
['3', 'Pineapple', '40000', '44', 'HR']
A:Add D:Delete O:Output V:ViewAll Q:Quit : O
HR Department Employees Age 40-45
3 Pineapple 40000 44 HR
A:Add D:Delete O:Output V:ViewAll Q:Quit : D
A:Add D:Delete O:Output V:ViewAll Q:Quit : V
['1', 'Apple', '5000', '32', 'Fruits']
['3', 'Pineapple', '40000', '44', 'HR']
A:Add D:Delete O:Output V:ViewAll Q:Quit : Q
```